

Tutorial 1 – Week 1

Imaginary and Complex Numbers

MTE 252

Niloo (Niloofar) Azad – Full TA

**MASc student in SYDE department
Vision and Image Processing Lab, EC4
Email: n2azad@uwaterloo.ca**

Office Hours and Tutorial Questions

I do not have fixed office hours. Office hours are available by request, so please email me with your questions, and we can arrange a meeting.

For tutorial sessions before quizzes and the midterm, please send me your questions a few days in advance.

If you would like me to go over specific assignment questions (or any other question) during a tutorial, please email me by Tuesday night.

Chapter BACKGROUND

Two forms of a complex number:

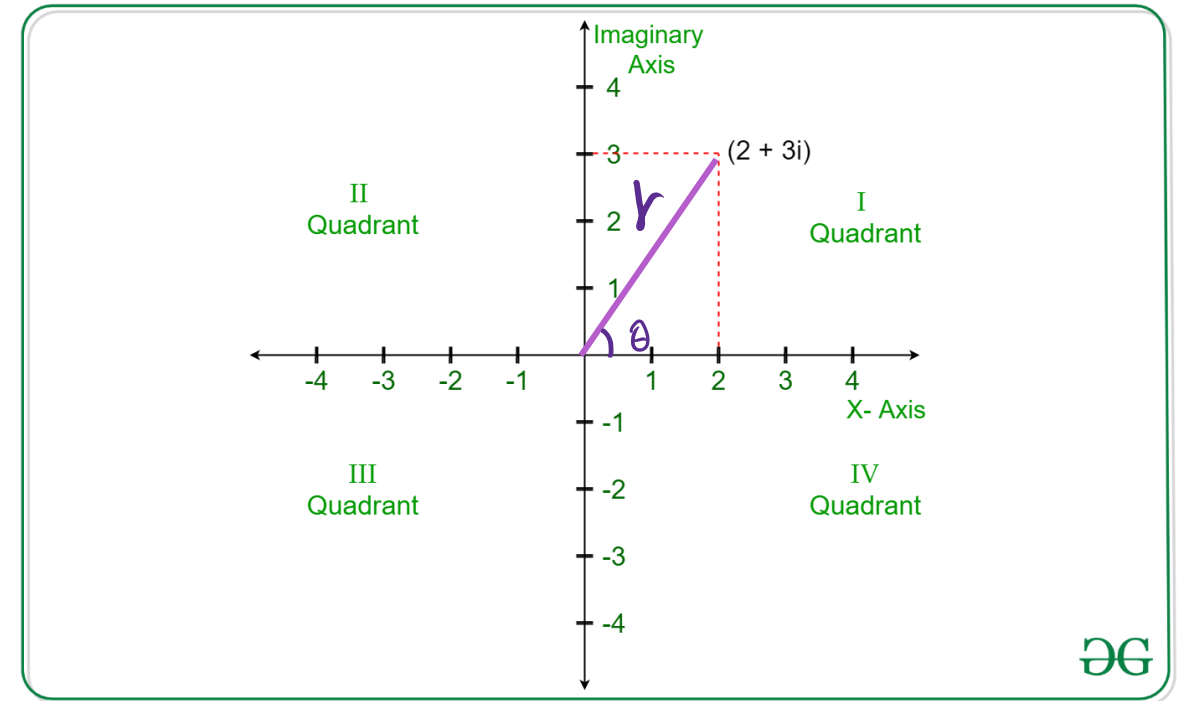
- Rectangular (Cartesian) form: $w = x + jy$
- Polar form: $w = re^{j\theta}$
 - Euler's formula: $e^{j\theta} = \cos(\theta) + j\sin(\theta)$

To convert rectangular form to polar form:

$$\begin{aligned} \text{(distance to origin)} \quad r &= \sqrt{x^2 + y^2} \\ \theta &= \tan^{-1} \left(\frac{y}{x} \right) \end{aligned}$$

Always check the quadrant:

- $x > 0, y > 0 \rightarrow \text{Quadrant I} \rightarrow \theta = \alpha$
- $x < 0, y > 0 \rightarrow \text{Quadrant II} \rightarrow \theta = \pi - \alpha$
- $x < 0, y < 0 \rightarrow \text{Quadrant III} \rightarrow \theta = -\pi + \alpha$
- $x > 0, y < 0 \rightarrow \text{Quadrant IV} \rightarrow \theta = -\alpha$



Question B.1-2

Express the following numbers in polar form:

(a) $w_a = 1 + j$

(b) $w_b = 1 + e^j$

(g) $w_g = (1 + j)/(-4 + j3)$

Solution B.1-2

Express the following numbers in polar form:

$$(a) \ w_a = \underbrace{1}_x + j \underbrace{1}_y$$

$$r = \sqrt{x^2 + y^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

$$\text{Polar form} \Rightarrow w_a = r e^{j\theta} \rightarrow \boxed{\sqrt{2} e^{j\frac{\pi}{4}}}$$

Solution B.1-2

Euler's formula $\Rightarrow e^{j\theta} = \cos(\theta) + j\sin(\theta)$

$$(b) w_b = 1 + e^j$$

$$e^j = e^{jx1} \rightarrow \theta = 1$$

$$e^{jx1} = \cos(1) + j\sin(1)$$

$$\text{so } w_b = 1 + e^j = \underbrace{1 + \cos(1)}_{\text{Real}(x)} + j \underbrace{\sin(1)}_{\text{Imaginary}(y)}$$

$$r = \sqrt{x^2 + y^2} = \sqrt{(1 + \cos(1))^2 + \sin^2(1)} = \sqrt{1 + 2\cos(1) + \underbrace{\cos^2(1) + \sin^2(1)}_{\textcircled{1}}} = \sqrt{2 + 2\cos(1)}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{\sin(1)}{1 + \cos(1)}\right) = \tan^{-1}\left(\tan\left(\frac{1}{2}\right)\right) = \frac{1}{2}$$

$$\text{Polar form } \rightarrow w_b = r e^{j\theta} = \sqrt{2 + 2\cos(1)} e^{j/2}$$

Half Angle Formulas

- $\sin\left(\frac{A}{2}\right) = \pm \sqrt{\frac{1 - \cos A}{2}}$
- $\cos\left(\frac{A}{2}\right) = \pm \sqrt{\frac{1 + \cos A}{2}}$
- $\tan\left(\frac{A}{2}\right) = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}} = \frac{\sin A}{1 + \cos A}$

Pythagorean Identities

1. $\sin^2(x) + \cos^2(x) = 1$
2. $1 + \tan^2(x) = \sec^2(x)$
3. $1 + \cot^2(x) = \csc^2(x)$

Even-Odd Identities

- $\sin(-x) = -\sin(x)$
- $\cos(-x) = \cos(x)$
- $\tan(-x) = -\tan(x)$



Solution B.1-2

$$(g) \quad w_g = \underbrace{(1+j)}_a / \underbrace{(-4+j3)}_b$$

polar form $w_g \rightarrow \frac{\text{Polar form } a}{\text{Polar form } b} = \frac{\sqrt{2} e^{j\frac{\pi}{4}}}{5 e^{j2.498}} = \boxed{\frac{\sqrt{2}}{5} e^{-j1.713}}$

a) $1+j \rightarrow \begin{matrix} x=1 \\ y=1 \end{matrix} \quad r = \sqrt{x^2+y^2} = \sqrt{2} \quad \left. \begin{matrix} \\ \end{matrix} \right\} \rightarrow \text{Polar form} \rightarrow \boxed{\sqrt{2} e^{j\frac{\pi}{4}}}$

$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}(1) = \frac{\pi}{4}$

b) $-4+j3 \rightarrow \begin{matrix} x=-4 \\ y=3 \end{matrix} \left. \begin{matrix} \\ \end{matrix} \right\} \rightarrow x < 0, y > 0 \rightarrow \text{Quadrant II} \rightarrow \theta = \pi - \alpha$

α is the reference angle $\tan^{-1}\left(\frac{3}{4}\right)$

$r = \sqrt{x^2+y^2} = \sqrt{16+9} = 5$, $\theta = \tan^{-1}\left(-\frac{3}{4}\right) \Rightarrow \theta = \pi - \tan^{-1}\left(\frac{3}{4}\right) \approx 2.498$

$\left. \begin{matrix} \\ \end{matrix} \right\} \rightarrow \text{Polar form} \rightarrow \boxed{5 e^{j2.498}}$

Question B.1-3

Express the following numbers in Cartesian (rectangular) form:

(b) $w_b = 3e^{j\pi/4}$

(c) $w_c = 1/e^j$

(e) $w_e = e^{j\pi/4} + 2e^{-j\pi/4}$

Solution B.1-3

Express the following numbers in Cartesian (rectangular) form:

$$(b) \ w_b = 3e^{j\pi/4} \rightarrow \theta = \frac{\pi}{4}$$

$$\text{Use Euler's formula } \rightarrow e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$3 e^{j\frac{\pi}{4}} = 3 \left(\underbrace{\cos\left(\frac{\pi}{4}\right)}_{\frac{\sqrt{2}}{2}} + j \underbrace{\sin\left(\frac{\pi}{4}\right)}_{\frac{\sqrt{2}}{2}} \right)$$

$$\text{Cartesian form } w_b = \boxed{3 \frac{\sqrt{2}}{2} + j 3 \frac{\sqrt{2}}{2}}$$

Solution B.1-3

Express the following numbers in Cartesian (rectangular) form:

(c) $w_c = 1/e^j = e^{-j}$ use Euler's formula

$$\hookrightarrow e^{-j} = \cos(-1) + j \sin(-1)$$

$$w_c = e^{-j} = \cos(1) - j \sin(1)$$

$$w_c = 0.5403 - j0.8415$$

Pythagorean Identities

1. $\sin^2(x) + \cos^2(x) = 1$
2. $1 + \tan^2(x) = \sec^2(x)$
3. $1 + \cot^2(x) = \csc^2(x)$

Even-Odd Identities

- $\sin(-x) = -\sin(x)$
- $\cos(-x) = \cos(x)$
- $\tan(-x) = -\tan(x)$

Question B.1-3

Express the following numbers in Cartesian (rectangular) form:

(e) $w_e = \underbrace{e^{j\pi/4}}_a + \underbrace{2e^{-j\pi/4}}_b$ use Euler's formula $\Rightarrow e^{j\theta} = \cos(\theta) + j\sin(\theta)$

$$\begin{aligned} a) e^{j\frac{\pi}{4}} &= \cos\left(\frac{\pi}{4}\right) + j\sin\left(\frac{\pi}{4}\right) = \boxed{\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}} \\ b) 2e^{-j\frac{\pi}{4}} &= 2\left(\underbrace{\cos\left(-\frac{\pi}{4}\right)}_{\frac{\sqrt{2}}{2}} + j\underbrace{\sin\left(-\frac{\pi}{4}\right)}_{-\frac{\sqrt{2}}{2}}\right) = 2\left(\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}\right) = \boxed{\sqrt{2} - j\sqrt{2}} \end{aligned}$$

$\xrightarrow{+} \boxed{\frac{3\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}}$

Question B.1-7

Given $w = x - jy$, determine:

- (a) $\text{Re}(e^w)$
- (b) $\text{Im}(e^w)$

$$e^w = e^{x - jy} = e^x \times e^{-jy}$$

use Euler's formula $\theta = -y$

$$e^{-jy} = \cos(-y) + j \sin(-y)$$

$$e^{-jy} = \cos(y) - j \sin(y)$$

$$e^w = e^x \times e^{-jy} = e^x (\cos(y) - j \sin(y))$$

$$e^w = e^x \cos(y) - j e^x \sin(y) \rightarrow \text{Re}(e^w) = e^x \cos(y)$$

$$\rightarrow \text{Im}(e^w) = -e^x \sin(y)$$

Pythagorean Identities

1. $\sin^2(x) + \cos^2(x) = 1$
2. $1 + \tan^2(x) = \sec^2(x)$
3. $1 + \cot^2(x) = \csc^2(x)$

Even-Odd Identities

- $\sin(-x) = -\sin(x)$
- $\cos(-x) = \cos(x)$
- $\tan(-x) = -\tan(x)$

You Try Question B.1-9

Given $w_1 = 3 + j4$ and $w_2 = 2e^{j\pi/4}$.

w_1 is already in rectangular (Cartesian) form

(d) Express $w_1 + w_2$ in standard rectangular form.

$$w_2 = 2e^{j\frac{\pi}{4}} \xrightarrow{\text{use Euler's formula}} 2e^{j\frac{\pi}{4}} = 2 \left(\underbrace{\cos\left(\frac{\pi}{4}\right)}_{\frac{\sqrt{2}}{2}} + j \underbrace{\sin\left(\frac{\pi}{4}\right)}_{\frac{\sqrt{2}}{2}} \right) = \sqrt{2} + j\sqrt{2}$$

$$w_1 + w_2 = 3 + j4 + \sqrt{2} + j\sqrt{2} = (3 + \sqrt{2}) + j(4 + \sqrt{2}) = 4.414 + j5.414$$